

CISC 327/227 Assignment 1

Role Based Requirements:

Individual Role: Renter Customer

- The system shall allow users to view the current availability status of specific gear items remotely.
- The system shall process gear checkouts using only a customer's name and phone number.
- The system shall process rentals without requiring the creation of a password-protected user account.
- The system shall allow customers to rent all equipment without placing a monetary deposit hold.
- The system shall provide a clear record of borrowed items and their exact due dates at checkout.
- The system shall send clear due date reminders to the customer before the rental period expires.
- The system shall allow customers to extend an active rental period by one or two days.

Joint Role: Rental Clerk

- The system shall return search results for gear availability and current borrower information within seconds.
- The system shall allow staff to search for rental records using a customer's name.
- The system shall allow staff to search for inventory using the gear's name rather than its exact ID number.
- The system shall allow staff to complete a checkout process using only a few taps or minimal data entry fields.
- The system shall mandate that a customer provides a valid ID before renting high-value equipment.
- The system shall require a deposit hold to be collected and recorded before

checking out high-value equipment such as kayaks and bikes.

- The system shall allow staff to flag returned inventory as "out of service" if it is damaged.
- The system shall prevent items flagged as "out of service" from being checked out to a new customer.

Individual Role: Shop Owner

- The system shall track what rentals a customer currently has rented out
- The system shall be able to track what gear is currently checked out
- The system shall track what time checked out gear is due back on
- The system shall be able to be run and maintained without a dedicated IT / technical assistant. The clerk should be able to understand the system with a short training session at most
- The system shall alert me or an employee of when an item is approaching being overdue. Allowing us to choose / define how soon "approaching" is to the actual deadline and allowing us to choose whether to contact the customer depending on if they are a regular or not. Limiting the annoyance for trusted customer.
- The system shall track who previously had a rental checked out and the damage state of the item when they return it. The system should also be able to add condition reports on initial purchase of the item and when the item gets repaired. Allowing us to alert said customer if they returned damaged equipment, and collecting payment for said damages.
- The system shall be easily upgraded in the future in case needs / requirements change as the company grows / changes. Adding online booking would be something that should be added in the future.
- The system shall keep track of the most popular items, and keep track of which items are requested to be rented when we do not have the item currently available.
- The system shall produce reports for its tracked metrics (currently checked out gear, overdue items, most popular items, requested items)
- The system shall be cheap to run with no ongoing licensing fees, and be able to run on a local machine
- The system shall keep track of the name and contact info of customers, and record how often they are checking out items, the system should allow us to add a mark next to customers names if we recognize them as a regular / trusted customer
- Open issue: The owner is leaning towards the idea of collecting deposits on high value items (valued over a certain dollar amount) to reduce losses on stolen / lost items, there is a disagreement on this opinion with the staff and customers and no final decision has been made yet

Group Consolidated Requirements List + Completed Quality Check Worksheet

Source	Requirement	Conflict	Ambiguity	Completeness	Consistency	Testability	Evolvability	Notes/Resolution
Renter	The system shall allow users to view the current availability status of specific gear items remotely.	Pass	Fail	Pass	Pass	Pass	Pass	Fails ambiguity. Rewritten to: "...via a public-facing web interface."
Renter	The system shall process gear checkouts using only a customer's name and phone number.	Pass	Pass	Fail	Pass	Pass	Pass	Fails completeness. Rewritten to add: "...using the phone number as the unique identifier."
Renter	The system shall process rentals without requiring the creation of a password-protected user account.	Pass	Pass	Pass	Pass	Pass	Pass	Passes all attributes.
Renter	The system shall allow customers to rent all equipment without placing a monetary deposit hold.	Fail	Pass	Pass	Pass	Pass	Pass	Fails conflict. Renter opposes deposits, but Clerk demands them. Flagged as open decision.
Renter	The system shall provide a clear record of borrowed items and their exact due dates at checkout.	Pass	Fail	Pass	Pass	Pass	Pass	Fails ambiguity. Rewritten to: "The system shall generate a receipt listing borrowed items and due dates."
Renter	The system shall send clear due date reminders to the customer before the rental period expires.	Pass	Pass	Fail	Pass	Pass	Pass	Fails completeness. Rewritten to: "...send an SMS reminder 24 hours before the rental period expires."
Renter	The system shall allow customers to extend an active rental period by one or two days.	Pass	Pass	Fail	Pass	Pass	Pass	Fails completeness. Rewritten to: "...extend an active rental period by 1-2 days via staff intervention."
Clerk	The system shall return search results for gear availability and current borrower information within seconds.	Pass	Fail	Pass	Pass	Pass	Pass	Fails ambiguity. Rewritten to: "...within 3 seconds."
Clerk	The system shall allow staff to search for rental records using a customer's name.	Pass	Pass	Pass	Pass	Pass	Pass	Passes all attributes.
Clerk	The system shall allow staff to search for inventory using the gear's name rather than its exact ID number.	Pass	Pass	Pass	Pass	Pass	Pass	Passes all attributes.
Clerk	The system shall allow staff to complete a checkout process using only a few taps or minimal data entry fields.	Pass	Fail	Pass	Pass	Pass	Pass	Fails ambiguity. Rewritten to: "...using no more than 3 data entry screens."

Clerk	The system shall mandate that a customer provides a valid ID before renting high-value equipment.	Pass	Pass	Fail	Pass	Pass	Pass	Fails completeness. Rewritten to: "...equipment with a replacement value over \$100."
Clerk	The system shall require a deposit hold to be collected and recorded before checking out high-value equipment such as kayaks and bikes.	Fail	Pass	Fail	Pass	Pass	Pass	Fails conflict and completeness. Flagged as open conflict for Shop Owner.
Clerk	The system shall allow staff to flag returned inventory as "out of service" if it is damaged.	Pass	Pass	Pass	Pass	Pass	Pass	Passes all attributes.
Clerk	The system shall prevent items flagged as "out of service" from being checked out to a new customer.	Pass	Pass	Pass	Pass	Pass	Pass	Passes all attributes.
Owner	The system shall track what rentals a customer currently has rented out	Pass	Fail	Pass	Pass	Fail	Pass	"Track" doesn't define the output. Rewritten to "...allow staff to view active rentals associated with a customer."
Owner	The system shall be able to track what gear is currently checked out	Pass	Fail	Pass	Pass	Fail	Pass	Same Issue: can be merged into previous requirement "...allow staff to view active rentals, including rentals associated to customers."
Owner	The system shall track what time checked out gear is due back on	Pass	Fail	Pass	Fail	Pass	Pass	"Time" alone is unclear, should it include date and time? Current overdue rule is tracked by date not time, rewritten as "...record a due back date for each item at checkout."
Owner	The system shall be able to be run and maintained without a dedicated IT / technical assistant. The clerk should be able to understand the system with a short training session at most	Pass	Fail	Fail	Pass	Fail	Pass	Can be split into 2 requirements: one for maintenance and one for on boarding the program. "maintained without IT" and "short training" are unable to be measured. Easily maintainable and easy to run requirements swapped to an easily understandable GUI requirement below.

Owner	The system shall alert the owner or an employee of when an item is approaching being overdue. Allowing us to choose / define how soon "approaching" is to the actual deadline and allowing us to choose whether to contact the customer depending on if they are a regular or not. Limiting the annoyance for trusted customer.	Fail	Fail	Fail	Fail	Fail	Pass	Conflicts with the 24 hour SMS requirement, Alert doesn't specify how, no default overdue value is given, "limiting annoyance" for customers is not measurable. Issue can be split and resolved: dashboard alerts and reminder opt outs covered in final requirements
Owner	The system shall track who previously had a rental checked out and the damage state of the item when they return it. The system should also be able to add condition reports on initial purchase of the item and when the item gets repaired. Allowing us to alert said customer if they returned damaged equipment, and collecting payment for said damages.	Pass	Fail	Fail	Fail	Fail	Pass	Again is multiple requirements merged into one, split in final requirements list. Damage state is ambiguous, resolved later to define how the damaged state is recorded. Doesn't say how payment is collected. Partially merged with already existing requirements (condition reporting and out-of-service) split into R16, R18, R19, R20
Owner	The system shall be easily upgraded in the future in case needs / requirements change as the company grows / changes. Adding online booking would be something that should be added in the future.	Pass	Fail	Fail	Pass	Fail	Pass	"Easily" cannot be measured or tested. Online booking may conflict with the renters no accounts requirement, moved to future requirements rather than current final requirements list.
Owner	The system shall keep track of the most popular items, and keep track of which items are requested to be rented when we do not have the item currently available.	Pass	Fail	Fail	Pass	Pass	Pass	"popular" is not a defined metric, how to keep track of requests also undefined. Merged into existing reporting requirements. Covered by reporting requirements
Owner	The system shall produce reports for its tracked metrics (currently checked out gear, overdue items, most popular items, requested items)	Pass	Fail	Fail	Pass	Pass	Pass	Reporting periods undefined, how reports will be handled / displayed left ambiguous, merged into existing report requirements

Owner	The system shall be cheap to run with no ongoing licensing fees, and be able to run on a local machine	Fail	Fail	Pass	Pass	Fail	Pass	"Cheap" is not explicitly measurable, Local Machine requirement can work with remote availability but needs setup which conflicts with the no IT requirement. Sent to ongoing issues.
Owner	The system shall keep track of the name and contact info of customers, and record how often they are checking out items, the system should allow us to add a mark next to customers names if we recognize them as a regular / trusted customer	Pass	Fail	Pass	Fail	Fail	Pass	"contact Info" left ambiguous, uses "should" and "we". Name, phone, and trusted regular tag already covered; rental frequency is new. Made into new requirement to purely track rental frequency.

AI Generated Requirements List + Completed Quality-Check Worksheet

Source	Requirement	Conflict	Ambiguity	Completeness	Consistency	Testability	Evolvability	Notes/Resolution
AI	The system shall maintain a record of all gear items, including a unique identifier, category, and current status.	Pass	Pass	Pass	Pass	Pass	Pass	Passes all attributes. Covered by inventory requirements
AI	The system shall allow staff to create a rental record linking a specific gear item to a renter.	Pass	Pass	Pass	Pass	Pass	Pass	Passes all attributes. Covered by checkout requirements
AI	The system shall capture the renter's name and contact information at the time of rental.	Pass	Pass	Fail	Pass	Pass	Pass	Fails completeness. Rewritten to specify "a valid phone number."
AI	The system shall record the checkout date and the due-back date for each rental.	Pass	Pass	Pass	Pass	Pass	Pass	Passes all attributes.
AI	The system shall record the actual return date when gear is checked back in.	Pass	Pass	Pass	Pass	Pass	Pass	Passes all attributes.
AI	The system shall display which gear items are currently overdue.	Pass	Pass	Fail	Pass	Pass	Pass	Fails completeness. Rewritten to: "...display a list of overdue items on the staff dashboard."

[illegible]

Comparison Write-Up

Human vs. AI Requirements Analysis

Comparing the human-generated stakeholder requirements with the AI-generated requirements revealed distinct differences in focus, specifically regarding business logic versus back end technical logic. The human role-play process was essential for surfacing real-world business constraints and stakeholder conflicts that the AI failed to anticipate. Conversely, the AI provided a highly structured baseline for database management and system state transitions that the human actors overlooked.

Where the Lists Agreed

The AI and human lists both included searching rentals by customer name, recording due back dates, flagging overdue rentals, taking note of damaged gear, and blocking checkout of unavailable items.

What the AI Missed

The AI generated a generic inventory system and completely missed the nuanced business operations of a rental shop. Most notably, it failed to identify the core stakeholder conflict regarding monetary deposits, where the Renter opposed deposits to reduce friction, but the Clerk demanded them to secure high-value inventory. The AI also missed usability and operational constraints, such as the Clerk's need for a rapid checkout process (under 3 seconds and minimal screens) and the Owner's requirement for a graphical user interface to avoid command-line administration.

What the AI Added

While the human requirements focused heavily on user experience and business rules, the AI excelled at defining the record keeping logic necessary for the system to function securely. The AI introduced critical technical requirements, including as generating unique rental transaction IDs, time stamping all events, and implementing state-locking mechanisms to prevent items marked as "rented" from being checked out concurrently.

Conclusion

Neither list was sufficient on its own. The final refined requirements list required synthesizing both sources: treating the AI's list as the technical framework and using the human-generated list to enforce business rules, dictate user interfaces, and resolve operational conflicts.

Final Refined Requirements List

Inventory

- **R1.** The system shall maintain a record of all gear items, including a unique identifier, category, current status, and replacement value.
- **R2.** The system shall allow staff to add, edit, and remove gear items.

Checkout

- **R3.** The system shall allow users to view the current availability status of specific gear items via a public-facing web interface.
- **R4.** The public-facing web interface shall display only gear availability and shall not provide access to customer records or staff functions.
- **R5.** The system shall process gear checkouts using only a customer's name and phone number, using the phone number as the unique identifier.
- **R6.** The system shall process rentals without requiring the creation of a password-protected user account.
- **R7.** The system shall mandate that a customer provides a valid ID before renting equipment with a replacement value over \$100.
- **R8.** The system shall require a deposit hold to be collected and recorded before checking out equipment with a replacement value over \$100.
- **R9.** The system shall allow staff to complete a checkout process using no more than 3 data entry screens.
- **R10.** The system shall generate a unique rental transaction ID for each checkout event.
- **R11.** The system shall record the checkout date and the due-back date for each rental.
- **R12.** The system shall generate a receipt listing borrowed items and due dates.
- **R13.** The system shall prevent a gear item from being checked out if its status is already "rented."
- **R14.** The system shall prevent items flagged as "out of service" from being checked out to a new customer.

Returns and Damage Conditions

- **R15.** The system shall record the actual return date when gear is checked back in.
- **R16.** The system shall require staff to record the item's condition when gear is checked back in.
- **R17.** The system shall allow staff to flag returned inventory as "out of service" if it is damaged.
- **R18.** The system shall allow staff to add condition reports to a gear item's record, including on initial purchase and after repair.
- **R19.** The system shall allow staff to look up the rental history of a specific gear item, including who last rented it.
- **R20.** The system shall allow staff to record a damage charge against a rental, apply it to the deposit hold where one exists, and notify the renter by SMS or email.
- **R21.** The system shall release the deposit hold when gear is checked back in, unless a damage charge is recorded.
- **R22.** The system shall timestamp all rental and return events.

Due Dates / Reminders

- **R23.** The system shall allow customers to extend an active rental period by 1-2 days via staff intervention.
- **R24.** The system shall flag a rental as overdue automatically when the current date exceeds the due-back date.
- **R25.** The system shall display a list of overdue items on the staff dashboard.
- **R26.** The system shall alert staff on the dashboard when a rental is approaching its due-back date.
- **R27.** The system shall send an SMS reminder 24 hours before the rental period expires.

- **R28.** The system shall allow staff to change the advance period used for R25 and R26 (default: 24 hours).
- **R29.** The system shall allow staff to disable automatic reminders for customers tagged as Trusted Regular.

Customers and Search

- **R30.** The system shall allow staff to toggle a Trusted Regular tag on a customer's record.
- **R31.** The system shall display the total number of past rentals on each customer's record.
- **R32.** The system shall allow staff to view active rentals, including rentals associated to customers.
- **R33.** The system shall allow staff to search for rental records using a customer's name.
- **R34.** The system shall allow staff to search for inventory using the gear's name rather than its exact ID number.
- **R35.** The system shall return search results for gear availability and current borrower information within 3 seconds.

Reporting

- **R36.** The system shall generate a report of currently checked-out gear and overdue rentals.
- **R37.** The system shall generate a monthly report of the most frequently rented items.
- **R38.** The system shall allow staff to log a customer request for gear that is currently unavailable.
- **R39.** The system shall generate a monthly report of logged requests for unavailable gear.

System Constraints

- **R40.** The system shall allow all administrative tasks to be performed through a graphical interface, with no command-line access required.
- **R41.** A new staff member shall be able to complete a checkout unassisted after no more than 30 minutes of training.
- **R42.** The system shall operate without recurring licensing or subscription fees.
- **R43.** The system shall run on a single on-premises computer.

Future Considerations (Not part of current final requirements)

- Online booking: Conflicts with current no account requirements, Owner mentioned it would be a nice addition in the future

Open Issues:

- Public availability page with owners requirement of the system running on a on-premises computer, without dedicated IT staff it would be difficult to keep the system secure, conflicting with Owners requirement of no on site IT.

Pre-Work Checklist

- Install a code editor (VS Code recommended).
- Install Git and confirm it works by running: `git --version`
- Create a GitHub account or use already created one. This is the account you will use for
- the whole course.
- Create the group's GitHub repository. One member should create it and add the rest of the
- group as collaborators.
- Invite your assigned TA as a collaborator on the repository.
- Confirm that every group member can clone the repository

Evidence:

<https://github.com/Braydenp006/CISC327A1/settings>

